TryHackMe — Computer Fundamentals Module

Windows Basics

Learning Notes — Desktop, File Explorer, Settings, Task Manager & Windows Security

> ***"Every operating system has its own personality, and Windows is the most familiar one: the world's most widely used desktop operating system."***

Desktop & Taskbar	File Explorer	Settings	Task Manager	Windows Security	Lab Answers

1. Introduction — Scenario & Windows History

In this room, you take on the role of a new employee starting your first day at TryHatMe. You log in to your Windows Server 2019 workstation, navigate the desktop, manage files, configure settings, monitor performance with Task Manager, and review security protections. Everything is explored through practical, hands-on tasks that mirror real daily Windows usage.

1.1 A Brief History of Windows

Year	Milestone
1981	MS-DOS released — a text-only black-screen OS. Users typed commands instead of clicking icons.
1985	Windows 1.0 released — a basic GUI shell built on top of MS-DOS. Introduced windows, menus, and mouse support.
1990s	Windows 3.1, 95, 98 — rapid GUI evolution. Windows 95 introduced the Start menu and taskbar (still present today).
2001	Windows XP — a major milestone combining consumer and professional versions. Widely considered the most popular Windows ever.
2007	Windows Vista — introduced UAC (User Account Control) for privilege management (controversial for excessive prompts).

2009	Windows 7 — refined Vista's approach; widely praised. UAC improved to be less intrusive.
2015	Windows 10 — returned the Start menu (removed in Windows 8), introduced virtual desktops and improved security.
2019	Windows Server 2019 — the version used in this room's lab VM. Enterprise-grade server OS with Desktop Experience.
2021	Windows 11 — refreshed UI, new Start menu position, enhanced security requirements (TPM 2.0 mandatory).

Why Windows for Security?

Windows is the most widely deployed desktop OS globally (~72% market share). This means the vast majority of enterprise endpoints, corporate workstations, and servers run Windows. Understanding Windows is not optional for security professionals — it is where most attacks happen, most investigations occur, and most defensive tooling operates.

2. Authentication & Windows Account Types

Before gaining access to the Windows Desktop, you must authenticate — prove your identity to the system. The login screen accepts a password, PIN, or other verification method. Once authenticated, the OS determines what you are allowed to do based on your account type.

Guest	Standard	Administrator
Restricted account for temporary access. Minimal permissions. Cannot change system settings or install software. Intended for short-term visitors.	Everyday user account. Can run applications and change personal settings. Cannot install software system-wide or change security settings.	Full control over the system. Can install software, change configurations, manage users, and access all files. Used in this room's lab.

Lab Note — Administrator Account

The room automatically logs you in as an Administrator on the Windows Server 2019 VM. This allows you to explore all system settings, install programs, and perform administrative actions without permission restrictions. In real enterprise environments, day-to-day work should use a Standard account — Administrator access only when needed (principle of least privilege).

2.1 UAC — User Account Control

UAC (User Account Control) is Windows' privilege escalation prompt system. When a Standard user or even an Administrator attempts an action that requires elevated privileges (installing software, changing system settings), Windows displays a UAC prompt requesting confirmation. This prevents malware and accidental changes from silently making system-wide modifications.

- Standard users see a credential prompt — must enter an Admin username and password.
- Administrators see a consent prompt — just click Yes/No to approve or deny.
- UAC was introduced in Windows Vista and refined in Windows 7 onwards.
- Security significance: attackers seek UAC bypass techniques to escalate privileges without triggering the prompt.

3. The Windows Desktop & Interface

The Windows Desktop is the main workspace — the first area you see after login. Using the airport analogy: if the login screen is the security checkpoint, the Desktop is the airport terminal. All other areas branch out from here.

3.1 Desktop Components

Component	What It Does
Desktop Icons	Shortcuts to frequently used files, folders, and applications. Fully customisable — right-click the desktop to add/remove icons. The Recycle Bin lives here.
Start Menu	Primary access point for all applications, settings, and power options. Click the Windows icon (bottom-left taskbar). Airport analogy: the departure board — shows what is available.
Search Bar	Quickly find applications, files, folders, and system settings by typing keywords. Accessible from the taskbar or Start menu.
Task View	Shows all currently open windows and virtual desktops. Click to quickly switch between windows or create a new virtual desktop.
Pinned Apps/Folders	Frequently used applications and folders pinned to the taskbar for one-click access — File Explorer, browser, etc.
System Tray	Bottom-right area: network status, audio volume, date/time, battery (on laptops), notification badges.
Notifications	Bell icon — displays system and application notifications. Also provides quick access to network settings.
Date & Time	Click the clock to open the full calendar. Right-click to open date/time settings.

Built-in Windows Tools

Windows ships with powerful built-in tools available immediately via the Start menu or Search: Notepad (text editing), File Explorer (file management), Task Manager (process/performance monitoring), Settings (system configuration), Windows Security (antivirus/firewall), Command Prompt & PowerShell (CLI), and many more. These are like airport services — already available in the terminal.

4. File Explorer & Windows File System

File Explorer — Navigate, manage, and organise files and folders

How to open: Taskbar icon (folder icon) | Start Menu → File Explorer | Win + E shortcut

- Windows uses a hierarchical folder structure — folders contain subfolders and files, organised in a tree.
- Common top-level locations: Desktop, Documents, Downloads, Pictures, Videos, OneDrive.
- The address bar shows the full path to the current folder: C:\Users\Administrator\Desktop\TryHatMe Onboarding
- Left panel (Navigation pane): quick access to This PC, network locations, and pinned folders.
- Right panel (Content area): shows files and folders in the current directory.
- Built-in search: search bar in the top-right — searches within the current folder and subfolders.
- File operations: copy (Ctrl+C), cut (Ctrl+X), paste (Ctrl+V), rename (F2), delete (Del), new folder (Ctrl+Shift+N).
- View options: Details view shows file size, type, and date modified — useful for forensics.

Security relevance: File Explorer access control is enforced by NTFS permissions. Folders with sensitive data should have restricted permissions. Attackers navigating a compromised system use File Explorer (or CLI) to locate credentials, configuration files, and sensitive data. Temporary files (%TEMP%), AppData, and ProgramData are common malware hiding spots.

4.1 Windows File Paths

Windows uses backslashes (\) to separate path components — unlike Linux/macOS which use forward slashes (/). Understanding file paths is essential for navigating the system both in the GUI and in security tools.

```
# Windows file path format:
C:\Users\Administrator\Desktop\TryHatMe Onboarding\Welcome.txt
#   ^    ^                  ^         ^                    ^
#   |    |                  |         |                    └── filename
#   |    |                  |         └─────────────────────── subfolder
#   |    |                  └───────────────────────────────── Desktop
#   |    └──────────────────────────────────────────────────── User profile
#   └───────────────────────────────────────────────────────── Drive letter
(C:)
```

```
# Key Windows paths:
C:\Windows\System32            # Core OS files and system executables
C:\Users\<username>            # User's profile folder
C:\Users\<username>\AppData # Hidden folder — app config and data
C:\Program Files               # 64-bit installed applications
C:\Program Files (x86)         # 32-bit installed applications
C:\Temp or %TEMP%              # Temporary files (malware often uses this)
```

4.2 NTFS — The Windows File System

NTFS (New Technology File System) is the standard file system for modern Windows. It provides the features that make Windows both functional and secure:

NTFS Feature	What It Provides
Permissions	Access Control Lists (ACLs) — fine-grained control over who can read, write, or execute each file or folder, down to the individual file level.
Encryption (EFS)	Encrypting File System — individual files and folders can be encrypted so only the owner can read them.
Journaling	NTFS keeps a transaction log — if the system crashes mid-write, the file system can recover without corruption.
Compression	Files and folders can be compressed to save disk space transparently.
Timestamps	NTFS tracks four timestamps per file: Modified, Accessed, Created, Entry Modified (MFT entry). Critical for digital forensics.
Alternate Data Streams (ADS)	Hidden data streams within NTFS files — can be used by malware to hide data inside legitimate files without visible size changes.

Lab Task — TryHatMe Onboarding Folder

Open the TryHatMe Onboarding folder on the Desktop in File Explorer. The Welcome.txt file inside contains the first flag: THM{welcome_to_tryhackme}. The path is: C:\Users\Administrator\Desktop\TryHatMe Onboarding\Welcome.txt. Then use the TryHatMeWelcome installer in the same folder to install the application and retrieve the installation flag.

5. Windows Settings App

Windows Settings — Configure and personalise your Windows environment

How to open: Start Menu → Settings (gear icon) | Win + I | Search 'Settings'

- The Settings app is the modern replacement for the legacy Control Panel — centralises all system configuration.
- System: Display, Sound, Notifications, Storage, About your PC (device info, RAM, OS version).
- Devices: Bluetooth, printers, mouse, keyboard, touchpad configuration.
- Network & Internet: Wi-Fi, Ethernet, VPN, proxy, firewall profile status.
- Personalisation: Desktop background, colours, lock screen, themes, Start menu layout.
- Apps: Installed applications, default apps, app permissions, startup programs.
- Accounts: User accounts, login options (PIN, password, Windows Hello biometrics), family settings.
- Time & Language: Date, time, time zone, region, language, keyboard layout.
- Update & Security: Windows Update, Windows Security (antivirus/firewall), backup, recovery, developer options.

Security relevance: Settings app security sections: Windows Security (antivirus, firewall), Windows Update (patching), Account management (MFA, sign-in options). Attackers with system access use Settings to create new accounts, disable security features, or change firewall settings. Audit logs capture Settings changes for incident response.

5.1 About Your PC — System Information

The 'About your PC' page (Settings → System → About) provides a complete overview of the device's hardware and OS configuration. This is the first place to check when investigating an unfamiliar machine:

Section	Information Shown	Lab Answer
Device specifications	Device name, processor, installed RAM, system type (32/64-bit)	`Device name: TryHatMe | RAM: 4.00 GB`
Windows specifications	Windows edition, version, OS build, installation date	`Version: 1809 (Windows Server 2019)`
Security	Secure Boot, TPM status, BitLocker encryption status	`(check in lab)`

5.2 Time & Language Settings

The Time & Language section of Settings shows the current date, time, time zone, and the configured country or region. This is tested in the lab:

- Navigate to Settings → Time & Language to find the country/region setting.
- Time zone awareness is important in security — log timestamps must be correctly aligned to UTC for accurate incident timelines.
- Region settings affect date formats — dd/mm/yyyy (UK) vs mm/dd/yyyy (US) — potential for confusion in log analysis.

6. Task Manager

Task Manager — Monitor processes, performance, startup items, and users

How to open: Ctrl + Shift + Esc | Right-click Taskbar → Task Manager | Ctrl + Alt + Del → Task Manager

- Processes tab: shows all running applications and background processes — name, CPU%, RAM usage, disk I/O, network usage.
- Performance tab: real-time graphs for CPU speed/utilisation, Memory (RAM used/available), Disk, Network, and GPU.
- App history tab: historical resource usage per application.
- Startup tab: lists programs that automatically start when Windows boots — and their startup impact (High/Medium/Low). Disable unwanted startup items here.
- Users tab: shows logged-in users and their resource consumption.
- Details tab: advanced view of all running processes with PIDs (Process IDs), status, and memory.
- Services tab: shows Windows services (background system processes) and their running/stopped status.
- Lab task: Performance tab → CPU section shows the CPU speed in GHz.

Security relevance: Task Manager is the first tool to open during a security investigation on a Windows machine. Look for: processes with unusual names or paths, processes running from %TEMP% or AppData, high CPU/memory usage by unknown processes, processes mimicking legitimate names (svchost.exe vs. svch0st.exe). Malware often disguises itself as a legitimate Windows process name.

7. Windows Security

Windows Security — Integrated antivirus, firewall, and security health dashboard

How to open: Start Menu → Windows Security | Settings → Update & Security → Windows Security | Search 'Windows Security'

- Windows Security is the built-in security dashboard — a single interface for all Windows security features.
- Virus & Threat Protection: real-time antivirus scanning powered by Microsoft Defender Antivirus.
- Firewall & Network Protection: Windows Defender Firewall status for each network profile.
- App & Browser Control: SmartScreen settings for app/file/browser protection.
- Device Security: hardware security features — TPM, Secure Boot, core isolation/memory integrity.
- Device Performance & Health: reports on storage capacity, apps/software, Windows Time service.
- Family Options: parental controls and device usage settings.

- Quick scan vs Custom scan: Quick scan checks common locations; Custom scan lets you choose specific folders.

Security relevance: Windows Security is the first line of defence on a Windows endpoint. Attackers commonly attempt to disable Defender (using PowerShell commands, registry changes, or AMSI bypass techniques) before deploying malware. Event ID 5001 in Windows Event Log records when real-time protection is disabled — a critical alert indicator.

7.1 Windows Defender Firewall — Network Profiles

Windows Defender Firewall manages inbound and outbound network traffic based on rules. It uses network profiles to apply different rules depending on the type of network the device is connected to:

Profile	When Used	Security Posture
Private	Trusted networks such as home or lab environments — automatically assigned when you mark a network as Private.	Less restrictive — allows more inbound connections for file sharing, printers, and local services.
Public	Untrusted networks — public Wi-Fi in cafes, airports, hotels. Default profile for unknown networks.	Most restrictive — blocks most inbound connections to protect from other users on the same public network.
Domain	Corporate/enterprise domain-joined networks — Active Directory managed environments.	Controlled by Group Policy — IT administrators define the rules centrally for all domain machines.

7.2 Lab Task — EICAR Test File Scan

The room includes a Windows Security practical task where you run a custom scan on a folder containing the EICAR test file — a safe, standard test file used to verify that antivirus software is working correctly without using real malware.

- Open Windows Security → Virus & Threat Protection → Scan options → Custom scan.
- Select the folder containing the EICAR test file and run the scan.
- After the scan, Windows Defender detects Virus:DOS/EICAR_Test_File.
- Click the detection and select 'See details' to view the Affected items section.
- The Affected items section shows the full file path and name of the detected file.

What Is the EICAR Test File?

The EICAR test file is an industry-standard benign file used to test antivirus detection without using real malware. Every antivirus product should detect it as 'EICAR-Test-Virus' or similar. If Windows Security detects the EICAR test file, your antivirus is functioning correctly. It is completely harmless — it is just a text file with a specific byte sequence that all AV vendors agree to detect.

8. Installing Software — The TryHatMeWelcome App

The room includes a practical task where you install the TryHatMeWelcome application from the TryHatMe Onboarding folder. This demonstrates the standard Windows software installation process and rewards you with a flag on completion.

Step	Action	Detail
1	**Navigate to the folder**	Open the TryHatMe Onboarding folder on the Desktop using File Explorer.
2	**Find the installer**	Locate the TryHatMeWelcome installer file inside the folder.
3	**Run the installer**	Double-click the installer. A UAC prompt may appear requesting Administrator confirmation.
4	**Complete installation**	Follow the installation wizard steps — Next, Next, Finish.
5	**Run the application**	Launch the installed application from the Desktop or Start Menu.
6	**Capture the flag**	The application displays a flag value on screen upon successful installation and launch.

9. Room Answers — Quick Reference

Question	Answer
What is the Device name specified in Device specifications?	`TryHatMe`
How much RAM is installed on your new work PC?	`4.00 GB`
Which Version of Windows Server 2019 Datacenter is installed?	`1809`
What is the flag value found within Welcome.txt?	`THM{welcome_to_tryhackme}`
What is the flag value after installing and running TryHatMeWelcome?	`(run the app to get flag)`
Which country or region is set in Time & Language settings?	`(check in lab)`
What is the speed of your CPU? (Task Manager → Performance tab)	`(check in lab)`
What is the file name shown in Affected items after EICAR scan?	`(check in lab)`

Which firewall profile is for trusted home/lab environments?	`Private`
Which firewall profile is for untrusted public Wi-Fi networks?	`Public`

10. Security Implications

Administrator Account = Full System Control

The Administrator account used in this lab has unrestricted access to every file, setting, and capability on the system. In real enterprise environments, day-to-day tasks should always use a Standard account — only elevate to Administrator when explicitly required. Malware running under an Administrator account can install drivers, modify security settings, and access all data without restriction.

File Paths Reveal the Playbook

Common Windows paths are exactly where attackers and defenders look first. %TEMP%, %AppData%\Roaming, C:\ProgramData, and C:\Windows\System32 are the most frequently used locations for malware staging, persistence, and hiding. Understanding normal vs. abnormal file locations is a core incident response skill.

NTFS Alternate Data Streams — Hidden Data

NTFS Alternate Data Streams (ADS) allow files to contain hidden secondary data streams. Malware uses ADS to hide payloads inside legitimate files — the file appears normal in File Explorer (no size change) but contains hidden executable code. The command 'dir /r' reveals ADS. This is a known evasion technique.

Task Manager Reveals Malicious Processes

The first action in any live Windows incident response is opening Task Manager. Look for: processes with misspelled names (svch0st.exe), processes running from unusual paths, multiple instances of processes that should only run once, and processes consuming unusual CPU/memory. The Details tab shows PIDs and command lines — critical for correlating with Event Logs.

Firewall Profiles Must Match Network Trust

If a Windows machine connects to a public Wi-Fi network but the firewall profile is set to Private, the machine allows inbound connections that should be blocked on untrusted networks. This is a common misconfiguration — always verify the firewall profile matches the network environment, especially on laptops used in multiple locations.

Defender Disabled = Open Season for Malware

One of the first things sophisticated malware does is disable Windows Defender. Attackers use PowerShell (Set-MpPreference -DisableRealtimeMonitoring $true), registry modifications, or process injection to evade Defender. Monitoring for Defender being disabled (Windows Event ID 5001) is a critical detection rule in any SIEM deployment.

11. Key Takeaways & Glossary

11.1 Key Takeaways

- Windows is the world's most widely deployed desktop OS — essential knowledge for every security professional.
- Three account types: Guest (minimal), Standard (everyday), Administrator (full control).
- UAC prompts protect against unauthorised system changes — attackers seek to bypass UAC for silent privilege escalation.
- The Desktop, Taskbar, Start Menu, and Search are the four primary navigation elements of the Windows interface.
- File Explorer uses backslash paths: C:\Users\Administrator\Desktop\folder\file.txt
- NTFS provides permissions, encryption (EFS), journaling, timestamps (MACE), and Alternate Data Streams.
- Lab: Device name = TryHatMe | RAM = 4.00 GB | Windows version = 1809 | Welcome.txt flag = THM{welcome_to_tryhackme}
- Settings → About your PC → Device specifications (RAM) and Windows specifications (version).
- Task Manager (Ctrl+Shift+Esc): Performance tab shows CPU speed; Details tab shows all processes with PIDs.
- Windows Security: antivirus + firewall + device health in one dashboard. EICAR test file tests AV detection.
- Firewall profiles: Private (trusted/home), Public (untrusted/Wi-Fi), Domain (enterprise/AD managed).

11.2 Core Glossary

Term	Definition
Desktop	The main workspace displayed after login — where files, folders, and shortcuts live.
Taskbar	The control strip at the bottom of the screen — Start menu, pinned apps, system tray, clock.
Start Menu	Primary access point for all applications, settings, and power options — click the Windows icon.
Administrator	The highest-privilege Windows account — full system control including software installation and user management.

Standard User	Everyday account with limited permissions — cannot make system-wide changes.
Guest	Minimal-permission temporary access account — cannot change settings or install software.
UAC	User Account Control — prompts for elevation when privileged actions are attempted. Prevents silent malware installation.
File Explorer	Windows' built-in file management tool — navigate folders, copy/move/delete files, view file paths.
File Path	The full address of a file: C:\Users\Administrator\Desktop\file.txt. Backslash separators.
NTFS	New Technology File System — the standard Windows file system. Supports permissions, encryption, journaling, and ADS.
ADS	Alternate Data Streams — hidden secondary data streams in NTFS files. Used by malware to hide payloads.
EFS	Encrypting File System — NTFS feature for encrypting individual files/folders so only the owner can read them.
Settings App	Modern Windows configuration centre — replaces Control Panel for most tasks.
About your PC	Settings section showing device name, RAM, Windows edition, and version (build number).
Task Manager	System monitoring tool — shows processes, CPU/RAM performance, startup items, and users. Ctrl+Shift+Esc.
PID	Process ID — unique number assigned to each running process by the OS. Used to identify and manage processes.
Windows Security	Built-in security dashboard — Microsoft Defender antivirus, Windows Firewall, device security.
Windows Defender	Microsoft's built-in antivirus and antimalware solution (part of Windows Security).
EICAR Test File	Industry-standard benign file to test antivirus detection. Detected as Virus:DOS/EICAR_Test_File.
Firewall Profile	Private (trusted), Public (untrusted), Domain (enterprise) — different inbound/outbound rules per network type.

11.3 Next Steps on TryHackMe

- **Immediate:** Continue with the Windows Fundamentals 1 room — deeper dive into the Windows GUI, file system, and tools.
- **Foundation:** Complete Windows Fundamentals 2 & 3 — system configuration, command line, Group Policy, and more.

- **Enterprise:** Study the Active Directory Basics room — Windows in enterprise environments, domain accounts, and Group Policy.
- **Security:** Take the Windows Privilege Escalation room — exploit misconfigurations to escalate from Standard to SYSTEM.
- **Forensics:** Study Windows Event Logs — the primary evidence source for Windows incident response.

Windows Is Where Security Work Happens

The majority of enterprise endpoints run Windows. The majority of malware targets Windows. Most credential theft, ransomware deployments, and lateral movement happens on Windows systems. Building deep Windows knowledge — the file system, the registry, processes, user management, and built-in security tools — is one of the highest-value skills a security professional can develop.

Source: TryHackMe — Windows Basics Room | tryhackme.com/room/windowsbasics